A Comparison of Surface and Intramuscular Myoelectric Signal Classification

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***Abstract*—The surface myoelectric signal (MES) has been used as an input to controllers for powered prostheses for many years. As a result of recent technological advances it is reasonable to assume that there will soon be implantable myoelectric sensors which will enable the internal MES to be used as input to these controllers. An internal MES measurement should have less muscular crosstalk allowing for more independent control sites. However, it remains unclear if this benefit outweighs the loss of the more global information contained in the surface MES. This paper compares the classification accuracy of six pattern recognition-based myoelectric controllers which use multi-channel surface MES as inputs to the same controllers which use multi-channel intramuscular MES as inputs. An experiment was designed during which surface and intramuscular MES were collected simultaneously for 10 different classes of isometric contraction. There was no significant difference in classification accuracy as a result of using the intramuscular MES measurement technique when compared to the surface MES measurement technique. Impressive classification accuracy (97%) could be achieved by optimally selecting only three channels of surface MES.**

***Index Terms*—Classification, EMG, intramuscular, myoelectric, pattern recognition, prostheses.**

I. Introduction

THE surface myoelectric signal (MES) has proven to be an effective control input for powered upper limb prostheses [1], [2]. Most commercially available systems employ a relatively simple scheme of encoding the amplitude of the MES, measured at one or more sites, to actuate one or more prosthetic devices such as an elbow, wrist or hand. Although these systems have been successful, they do not provide sufficient information to reliably control more than one function (or device) [3]; the extension to controlling multiple functions is a much more difficult problem. Unfortunately, these are the requirements of those with high-level (above the elbow) limb deficiencies, and these are the individuals who could stand to benefit most from a functional replacement of their absent limbs.

If one is to increase the number of devices under the control of the MES, it is clear that a more sophisticated means of discriminating different muscle states is needed. There are two major approaches that may be pursued to increase the accuracy of these controllers: 1) use *signal processing* to extract more information from the input signals; or 2) provide *more informative raw signals* to the controller.

A. Signal Processing

Sophisticated signal processing techniques employ pattern recognition to extract a wealth of control information from the surface MES. In the pattern recognition-based control paradigm, each movement class corresponds to a degree of freedom of prosthetic control. This idea is by no means new; indeed, the first pattern recognition-based control schemes were developed as early as the late 1960s and early 1970s [4]–[6]. Since then, dramatic improvements have been made in pattern recognition (feature extraction and classifiers), as well as in the speed, size and power consumption of embedded controllers. State-of-the-art methods in feature extraction and classifier design have been applied to pattern recognition-based MES control resulting in numerous different control schemes [7]–[17]. These methods have provided steady improvement in the accuracy of pattern recognition-based approaches, but it is important to consider what further improvement might be possible due to signal processing.

In the context of MES pattern classification, feature extraction attempts to extract usable information from the MES through development of feature sets which are chosen to preserve class separability. Choosing a feature set which is representative of the data is by no means trivial. In fact, it has been shown that classification accuracy is affected more by the choice of feature set than by the choice of classifier [1]. Pattern classification attempts to categorize measured data into a limited number of output classes. Most modern classification methods have been applied to MES control, including Bayesian classifiers [9], artificial neural networks [7], [8], [14], Gaussian mixture models [13], hidden Markov models [15], fuzzy logic [11], [12], [16], and genetic algorithms [17].

To illustrate the classification performance resulting from some selected feature sets and classifiers, consider the 6 class problem described in [13]. Data were collected from 12 normally limbed subjects by placing four equally spaced electrodes about the circumference of the forearm. Fig. 1 displays the result of a direct comparison of various feature sets; a sixth-order autoregressive (AR) model has been shown to provide the highest classification accuracy. Although the result shown in Fig. 1 was created by specifically considering a linear discriminant analysis (LDA) classifier, the results hold for all the classifiers shown in Fig. 2. Fig. 2 displays the result

Manuscript received November 15, 2006; revised October 7, 2006. This work was supported in part by NSERC Discovery Grant 171368-03 and Grant 217354-01, in part by the New Brunswick Foundation for Innovation, and in part by the Atlantic Innovation Fund. *Asterisk indicates corresponding author.*

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Digital Object Identifier 10.1109/TBME.2006.889192

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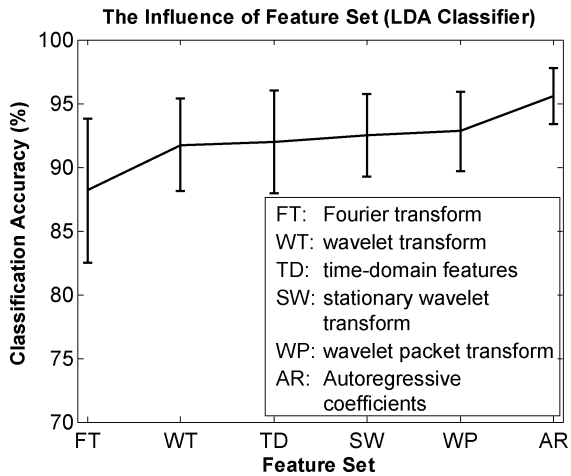

Fig. 1. The classification performance of a six-class control problem (wrist flexion, wrist extension, forearm supination, forearm pronation, hand open, and hand close), averaged over 12 normally limbed subjects. On these data, the AR feature set outperforms all other feature sets.

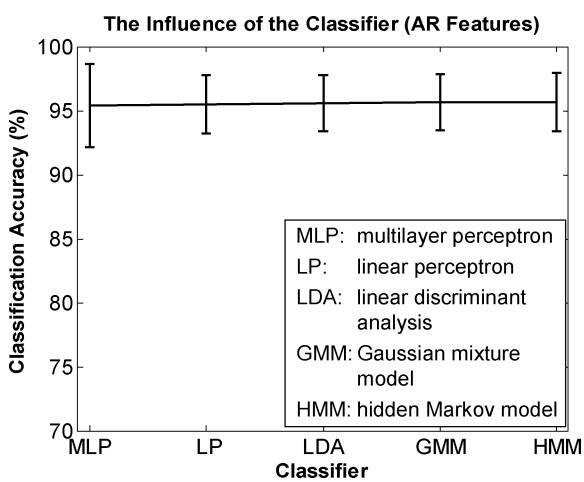

Fig. 2. A comparison of various pattern classifiers which was made using a sixth-order AR feature set. These data are the same as those described in Fig. 1. There is no significant difference between classifiers.

of a direct comparison of various pattern classifiers; they all result in essentially the same classification accuracy provided a representative feature set is used. The plot shown in Fig. 2 was created using the AR feature set; however the results are valid for all the feature sets shown in Fig. 1. These figures were created by merging the results presented in [13] and [15]. It is evident that very little difference separates the best-performing systems. This suggests that it is possible that an upper limit in accuracy has been reached that may be attributed solely to signal processing.

B. More Informative MES Data

In an effort to provide greater information content, it is natural to look to intramuscular recordings of MES activity. Myoelectric control systems typically employ bipolar surface electrodes. In terms of assessing the contribution of the active musculature to a given contraction pattern, the surface signal presents two distinct disadvantages. First, the contribution of the deeper muscles is obscured due to the tissue filtering effect and from the MES from more superficial muscles. Second, there may be a significant amount of crosstalk between muscles in the surface MES due to many factors, including the geometric arrangement of muscles, and the dispersive propagation of the signal in the volume conductor which is exacerbated by subcutaneous fat tissue signal [18].

Alternatively, the MES can also be measured invasively using needle or fine-wire electrodes. Clearly, this will reduce the tissue filter effect because there is no fat or skin between the signal source and the electrode allowing for a high quality MES measurement which is free from muscle crosstalk. To date it has not been practical to use invasive electrodes for the purpose of prosthetic control.

Weir *et al.* [19] are currently developing a new implantable myoelectric sensor (IMES) which transmits data through the skin using a wireless link. These IMES will be housed in BION II packaging capsules with a planned MES sample rate of 1.2 KHz. The capsules are small enough to permit injection through a 12 gauge hypodermic needle to a localized control site. A proof of concept system was built to demonstrate the technical feasibility of using reverse telemetry to transfer data from the electrode to receiver and forward telemetry to transmit power from a coil to the electrodes. Furthermore, a one subject fine wire intramuscular electrode experiment was performed which demonstrated acquisition of up to seven channels of independent intramuscular MES [19].

IMES would be particularly well suited for prosthetic limb control because the IMES could be placed in a control site while the power coil, data receiver, and control system could be mounted in the socket of the prosthesis. However, it remains unclear if the benefits of using intramuscular MES outweigh the loss of the more global information contained in the surface MES. This study compares the classification accuracy of several pattern recognition-based myoelectric controllers which use information extracted from surface MES to the same controllers which use information extracted from fine-wire intramuscular MES, which emulate the signals that would be provided by IMES devices.

II. Methodology

It was desirable to have a data set during which intramuscular and surface signals were collected simultaneously. Consequently, any variations in the manner the subjects performed the motions would be evident in both the surface and intramuscular data. The myoelectric control problem can be defined for any set of motions. It was decided to investigate a ten-class problem involving the degrees of freedom associated with the hand and wrist, as those with transradial limb deficiencies represent a large proportion of prosthetic users. The remainder of this section details the data collection process and the data processing parameters.

A. Choice of Motions and Intramuscular Control Sites

The 10 motions of interest in this study were associated with the three degrees of freedom of the wrist, two different hand grips and a rest state. In particular they were: forearm pronation, forearm supination, wrist flexion, wrist extension, radial deviation, ulnar deviation, key grip, chuck grip, hand open, and

a rest state. It has been shown that is it possible to obtain multiple intramuscular control sites in the forearm which should be sufficiently independent for multifunction control [19]. The internal control sites for this study were chosen to be: pronator teres and supinator teres to provide wrist rotation information, flexor digitorum sublimas and extensor digitorum communis to provide grip information, and flexor carpi ulnaris and extensor carpi ulnaris to give wrist flexion, extension, and deviation information.

B. Apparatus

Two separate systems were used to collect the MES; one for the surface MES and one for the intramuscular MES. The surface acquisition system consisted of a 16 electrode linear array (Clinical Prototype Manufactured by Laboratorio di Ingegneria del Sistema Neuromuscolare, Turin, Italy) with interelectrode spacing of 2 cm connected to a Prima EMG 16 amplifier (Prima Biomedical and Sport, Turin, Italy) configured to make differential measurements between adjacent electrodes. Each channel was filtered between 10–500 Hz using a fourth-order Bessel band pass filter with a gain of 2000 and a CMRR greater than 96 dB/channel. The 15-channel analog output of the amplifier was sent through a ribbon cable to a National Instruments DAQ-Card-6024E for 12–bit A/D conversion. The intramuscular MES acquisition system consisted of an intramuscular electrode connected to a custom built amplifier system and sampled by a Measurement Computing PCIM-DAS1602/16 16-bit A/D converter. The electrodes used were manufactured by Viasys Healthcare and consisted of 10 cm of insulated 44 gauge nickel alloy wires in a paired configuration which had an interelectrode spacing of 1 mm and were housed in a 27 gauge cannula. The signals were bandpass filtered between 10–3000 Hz and amplified with a gain of 2000. The amplifier used optical isolation for subject safety and had a rated CMRR of 100 dB.

One of the major difficulties of this experiment arose from the high number of channels of MES data which were collected; there were 15 channels of surface MES and six channels of intramuscular MES collected for each motion. Consequently, a choice was made to perform isometric contractions to prevent the motions from interfering with the lead wires from the electrodes. An arm brace which held the forearm stationary during the contractions was constructed. The arm was supported at the elbow and wrist by two pieces of wood covered with foam while the hand was inserted into a padded slot. During contractions the padded slot provided some resistance for the hand to push against, while the forearm was immobilized.

C. Protocol

The following experimental protocol was approved by University of New Brunswick's Research Ethics Board. Six normally limbed, healthy, male subjects between the ages of 23 and 30 were recruited to participate in the investigation. Informed consent was obtained from each subject. Next skin preparation was completed in the following manner: 1) the area around the circumference of the right forearm and a location just above the elbow on the back of the arm was rubbed lightly with an abrasive surface to remove the stratum corneum layer of the epidermis; 2) a conductive gel was rubbed over the same area

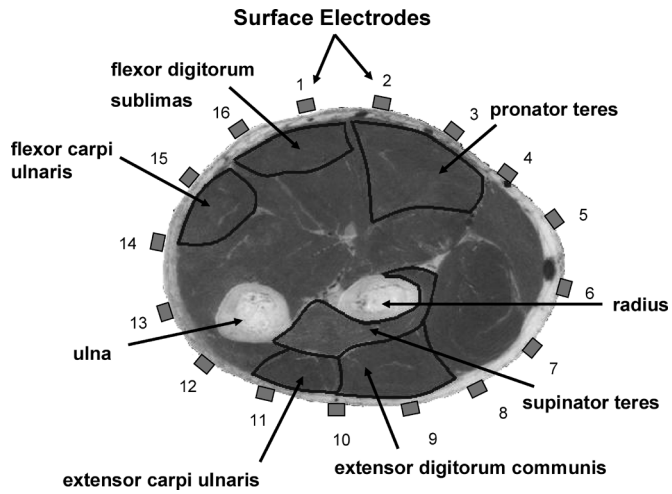

Fig. 3. A cross section of the upper forearm showing both the surface electrode locations and the internal control sites.

to reduce the skin/electrode resistance; 3) an alcohol wipe was used to remove any excess gel to prevent short-circuiting between adjacent electrodes; 4) a RedDot electrode was placed just above the elbow as a reference electrode. Following skin preparation, the intramuscular electrode insertions were completed by a physiatrist who had extensive experience localizing the muscles under investigation. The subject was instructed to complete various test contractions as the physiatrist located and inserted the electrodes into the muscles of interest. The subject was instructed to contract lightly as the insertion needle was withdrawn to better anchor the intramuscular electrodes in place. After all the intramuscular electrodes were inserted, the surface electrode array was applied to the forearm. The electrode array was simply wrapped around the circumference of the upper third of the forearm. Fig. 3 displays a cross section of forearm showing both the intramuscular and surface electrode locations. It should be noted that each subject's specific anatomy determined the precise locations of the surface channels with respect to the muscles shown in Fig. 3. However, an effort was made to consistently specify electrode 6 as located on the top of the forearm when the arm was placed inside the arm brace.

Following the application of all the electrodes, the arm was placed in the arm brace and connections were made to the amplifier units. Each surface and intramuscular channel was visually inspected to ensure that they contained valid MES. Specialized MATLAB-based acquisition software was written to guide subjects through a data acquisition session. Each session consisted of two trials of two repetitions of each motion. Subjects were prompted to complete medium force isometric contractions of 5–s duration followed by a brief rest period. The trials were separated by 2 min, and no subjects reported feeling fatigued.

D. Processing Parameters

Six different feature set/classifier combinations were used to make the comparison between the surface and intramuscular inputs. The time domain (TD) statistics described in [7], a sixth-order AR model plus the RMS signal value, and concatenated TD and sixth-order AR (TDAR) feature sets were used. The

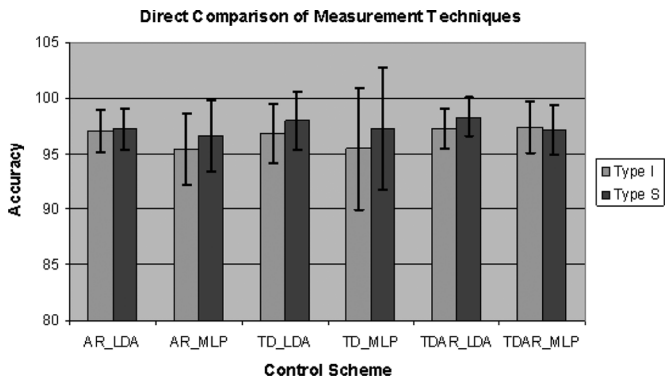

Fig. 4. The average classification accuracy and standard deviation for the six different control schemes. Type I indicates intramuscular measurements and Type S indicates surface measurements.

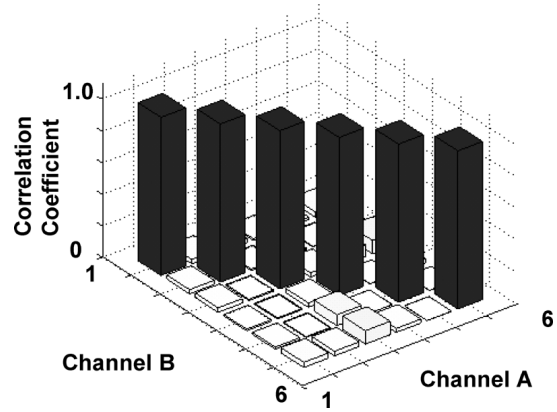

Fig. 5. The correlation coefficient between intramuscular MES channels. The values are averaged across all subjects.

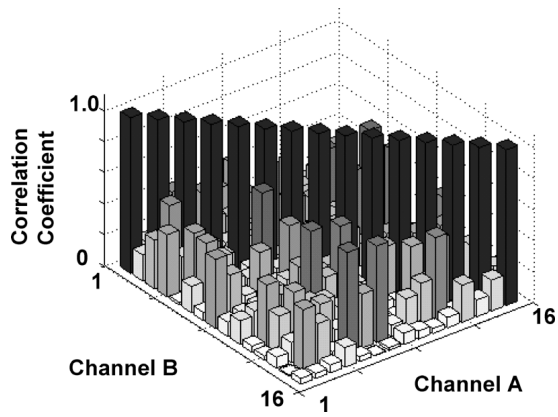

Fig. 6. The correlation coefficient between surface MES channels using a 16-channel array. The values are averaged across all subjects.

classifiers investigated were an LDA and a multilayer perceptron (MLP) artificial neural network with 12 hidden layer nodes which was trained using the backpropagation algorithm. For each combination, data analysis windows of 256 ms were used to extract features. The 256-ms data window was incremented by 32 ms, producing a decision at 32-ms intervals. These decisions were smoothed by determining the *majority vote* of the decisions comprising the current window, the previous eight windows, and the next eight windows [2]. This effectively looks ahead and behind by 256 ms, allowing 17 decisions to be used in a majority vote while keeping the user perceived delay less than 300 ms. The first 40 principal components of the feature sets were used in processing except when the feature vector had a dimension less than 40, in which case all principal components were used. The number of principal components was chosen empirically to be 40 principal components provided good results. Trial one of the data set was used to train each of the classifiers and trial two was tested to determine the classification accuracy of the control scheme.

III. Results

Fig. 4 displays the average classification accuracy and standard deviation across subjects of the six different feature set/classifier combinations. A paired T-test was used to test for a significant difference in classification accuracy resulting from the different measurement techniques for each of the control schemes investigated. The resulting p-values were all sufficiently large $(p > 0.10)$ to suggest that there was no significant difference between the measurement techniques; each control scheme yields very high classification accuracy.

It is impractical to consider embedding 15 surface electrodes in prostheses, at least using today's technology. Therefore, it is of interest to reduce the number of surface electrode channels without degrading classification performance. Also, the processing time it takes to form a decision is proportional to number of input channels. Six intramuscular channels would therefore require less than half of the processing time of the 15-channel surface electrode array. However, the global pickup and closely spaced channels of the surface electrode array results in some adjacent channels containing redundant information as compared to the internal channels which are generally uncorrelated. The correlation coefficients between channels of intramuscular and surface MES are displayed in Figs. 5 and 6, respectively. It is evident that the surface MES has higher correlation coefficients, indicating that channel crosstalk is present. It is possible that a subset of the 15 channels of surface data can be chosen while still maintaining high classification accuracy.

The TDAR feature set and LDA classifier were used to investigate how different channel subsets affected classification accuracy. Two methods were used to select channel subsets; a naïve method based on symmetric channel reduction (symmetrical method) and a brute force method which investigated all possible channel subsets (optimal method). In the symmetrical method, channel subsets were selected by simply choosing n equally spaced channels around the circumference of the forearm whereas in the optimal method, every possible combination of n channels was processed and the channels which provided the highest classification accuracy were selected as the channel subset. Fig. 7 displays classification accuracies resulting from using channel subsets of various sizes. In both cases it can easily be observed that as channels are added, the classification accuracy increases; however the increase is not linear, it quickly reaches a high value as the first few channels are added then slowly approaches a maximum. It can also been seen in Fig. 7 that the optimal channel subset method approaches the maximum more quickly than the symmetrical

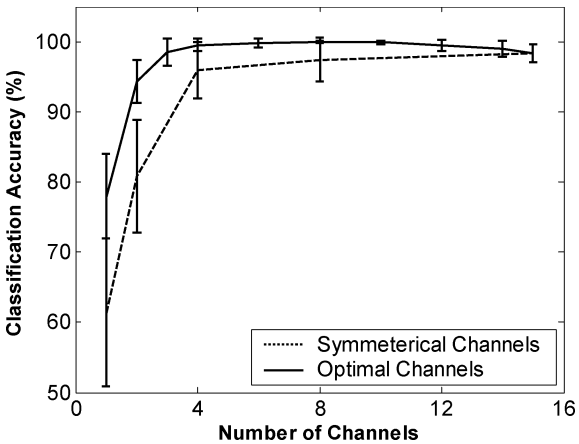

Fig. 7. The increase in classification accuracy as channels are added for both optimal and symmetrical channel subsets. The results shown here are for an LDA classifier with TDAR features. Similar results were obtained for all other feature set/classifier combinations.

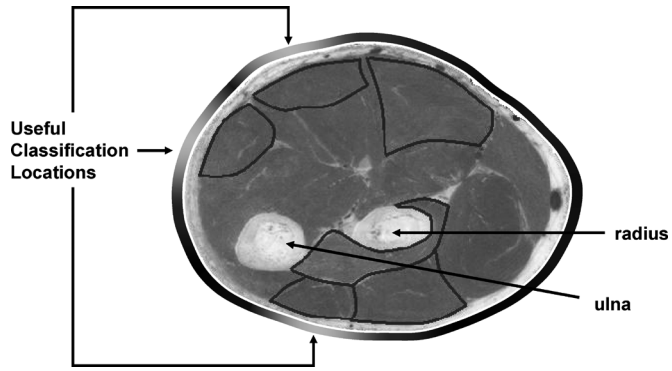

Fig. 8. Diagram showing the active areas of the forearm. The light regions of the ring surrounding the forearm indicate electrode placement areas which were most useful for classification.

method as one would intuitively expect. A two-way analysis of variance was used to determine that there was no benefit in adding more than four channels when using the symmetrical method or more than three channels when using optimal method. However, a slight drop in classification accuracy, significant at the $p = 0.05$ level, was observed when increasing from 8, 10, 12, or 14 optimal channels to 15 channels.

The optimal channel subsets were further investigated to identify the regions of the forearm where the optimal channels were located. Consequently, the optimal electrode placement locations were determined for this experiment. The following method was devised to determine which channels contributed the most information in discriminating amongst the 10 motions of interest.

- The classification accuracy was computed for all channel subsets for the $n = 2, 3, 4$ cases.
- A frequency count was kept of the actual channels used in the top five results for each case. Each time a channel was used as part of an optimal subset, the count was increased. The top five results were used because several channel subsets resulted in similar classification accuracies.
- The channels were mapped back to the corresponding locations on the forearm.

The results of mapping the channels back to the forearm are shown in Fig. 8. This suggests that surface electrodes placed over the extensors/supinator, flexor carpi ulnaris, and flexor digitorum subliminus are essential in providing good classification accuracy. Although the location of the optimal channels is not the same between all subjects, it should be noted that there was some variation in the manner the surface electrode was applied between subjects due to: 1) the need to insert around the intramuscular fine-wire electrodes; 2) the differences in the physical dimensions of the subjects forearm. A more judicious surface electrode placement strategy may result in more precise optimal channel locations. However, it can still be observed that there are general regions which may provide better discrimination information for the 10 motions investigated as shown in Fig. 8.

IV. Discussion

The results of the direct comparison between the surface and intramuscular measurement techniques suggest that the benefits associated with using intramuscular MES (localized signal, very little muscle crosstalk) do not outweigh the more global information contained in the surface MES for pattern recognition-based myoelectric control. Figs. 5 and 6 clearly indicate that the surface MES has higher inter-channel correlation than the intramuscular MES; however, the classification accuracy remains high for both cases. The implication is that pattern recognition-based myoelectric control systems can mitigate muscle crosstalk and that they possibly exploit spatial-temporal information between channels contained in the crosstalk for classification purposes. Consequently, strictly independent control sites are not a necessity, for pattern recognition-based myoelectric control. It is important to note that muscle crosstalk is not the only source of correlations in the data. Correlated interference sources such as 60-Hz power line interference also contributes to the correlation index. This is particularly evident in Fig. 6 where it is observed that some channels which are located on the opposite side of the arm have high correlation indices such as between channels 11 and 5. Upon closer inspection of the raw signals, it was observed that these channels had a higher 60-Hz noise component in addition to MES activity.

Although 15 channels of surface MES data were collected, only three channels are needed to provide excellent pattern recognition-based myoelectric control systems (97% classification accuracy on average), provided the channels are chosen carefully. When considering Fig. 7, it is interesting to note that there is a slight drop in classification accuracy when increasing from a 10-channel optimal subset to 15 channels. One possible explanation as to why the classification accuracy degrades when adding 14 or 15 channels is that there is some loss of useful data during the PCA; for the 15 channel input case, the dimensionality is being reduced from 165 to 40. Using more principal components may yield high classification accuracy; however this increased dimensionality of the input space would require more training data to provide a dense decision boundary to overcome the curse of dimensionality. A second possible

explanation is that adding additional channels to the data increases the complexity of the feature set. PCA assumes that the variance in the data corresponds to good feature clustering [20]. If adding additional channels to the data changes the feature cluster to a complex shape then performance of the PCA will be degraded. Clearly, more research needs to be conducted to determine the cause of classification degradation when using a high number channels. If PCA is not suitable for data with a high number channels then other data reduction techniques need to be investigated, otherwise training sets will need to be very large to overcome the curse of dimensionality.

It should be noted that only the classification accuracy of pattern recognition-based myoelectric control systems were considered in this work. Future research needs to be conducted to determine the suitability of intramuscular inputs for conventional myoelectric control systems. It is likely that the greater independence of the intramuscular signals will provide superior information allowing simultaneous, independent control of multiple DOF if the measurement sites are appropriately selected.

Finally, it should be noted that a limitation of the study is that the intramuscular MES signals examined, with the exception from the extensor carpi radialis, were measured from superficial muscles. From a control perspective, one advantage of using intramuscular electrodes is that deep muscles not obviously accessible through the surface EMG can be measured. This is not explicitly addressed in the study.

V. Conclusion

This paper compared the classification accuracy of several pattern recognition-based myoelectric controllers which use information extracted from surface MES to the same controllers which use information extracted from fine-wire intramuscular MES. In making this comparison, six different state-of-the-art pattern recognition-based myoelectric control systems were investigated. The results suggest that information extracted from either surface or intramuscular MES inputs yield excellent classification accuracy; however, there was no significant difference in classification accuracy for any of the six control schemes investigated for this particular experiment. Both measurement techniques yielded classification accuracies between 95% and 99%. Two methods of selecting surface channel subsets of various sizes were investigated. Using symmetrical channel subset selection, it was found that no significant increase in classification accuracy resulted from adding more than four channels. Using optimal channel subset selection, it was found that no significant increase in classification accuracy resulted from adding more than three channels. However, a slight drop in classification accuracy, significant at the $p = 0.05$ level, was observed when increasing from 8, 10, 12, or 14 channels to 15 channels. Furthermore, the optimal channel subsets were used to determine ideal electrode placement locations for the 10 motions under investigation. It is worth mentioning that only pattern recognition-based control systems were investigated in this work and the benefits of using intramuscular signals may be apparent when using conventional myoelectric control systems.

Acknowledgment

The authors would like to acknowledge the assistance of Dr. R. Leckey, who performed the intramuscular insertions, and B. Lock who developed the Matlab-based signal acquisition system.

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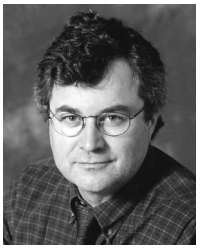

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